

Recall: Key Definitions

Hamiltonian cycle. A *Hamiltonian cycle* in a graph G is a cycle that contains *every* vertex of G (each vertex visited exactly once).

Unlike Eulerian tours (which cover every *edge*), Hamiltonian cycles cover every *vertex*. No efficient algorithm is known for deciding whether a graph has a Hamiltonian cycle.

Line graph. For a graph $G = (V, E)$, the *line graph* $L(G)$ has vertex set E (the edges of G), and two vertices $e, e' \in E$ are adjacent in $L(G)$ iff $e \cap e' \neq \emptyset$ (i.e. the edges share an endpoint in G).

Eulerian graph. A connected graph is *Eulerian* iff every vertex has even degree (Theorem 4.4.1).

Practice: Textbook Exercise 4.4.8 — Line graphs, connectivity, and Hamiltonian cycles

For a graph G , let $L(G)$ denote its line graph. Decide whether the following is true for every graph G :

- (a) G is connected if and only if $L(G)$ is connected.
- (b) G is Eulerian if and only if $L(G)$ has a Hamiltonian cycle.

Remarks

For part (a): Think about what a path in G corresponds to in $L(G)$, and vice versa. What happens when G has an isolated vertex?

For part (b): An Eulerian tour in G visits every *edge*; edges of G are *vertices* of $L(G)$. What does the Eulerian tour become in $L(G)$?

Answer: Part (a): FALSE in general (but true with a mild extra condition).

Counterexample. Let G consist of a single edge $\{u, v\}$ plus an isolated vertex w . Then G is *not* connected, but $L(G)$ has a single vertex (the edge $\{u, v\}$) and no edges, hence $L(G)$ is trivially connected. The issue is that isolated vertices of G are invisible to $L(G)$.

True direction ($\Rightarrow$ with no isolated vertices): Suppose G is connected with $|E| \geq 1$. Take any two edges $e, e' \in E(G)$. Since G is connected, there is a walk in G from an endpoint of e to an endpoint of e' . Each pair of consecutive edges in this walk shares a vertex, so they are adjacent in $L(G)$. This gives a walk from e to e' in $L(G)$, so $L(G)$ is connected.

Converse: If $L(G)$ is connected and G has no isolated vertices, then G is connected. (If G had two components each with ≥ 1 edge, no edge of one component shares an endpoint with an edge of the other, so $L(G)$ would be disconnected.)

Precise statement: If G has no isolated vertices and $|E(G)| \geq 1$, then G is connected $\iff L(G)$ is connected.

Part (b): TRUE (for graphs with $|E| \geq 1$).

($\Rightarrow$) Let G be Eulerian, so it has a closed Eulerian tour $e_1, e_2, \dots, e_m$ visiting every edge exactly once. Consecutive edges e_i and e_{i+1} share a vertex in G , so they are adjacent in $L(G)$. The sequence $e_1, e_2, \dots, e_m$ is therefore a cycle in $L(G)$ that visits every vertex of $L(G)$ exactly once — i.e. a Hamiltonian cycle in $L(G)$.

($\Leftarrow$) Suppose $L(G)$ has a Hamiltonian cycle $e_1, e_2, \dots, e_m, e_1$. Each consecutive pair e_i, e_{i+1} are adjacent in $L(G)$, meaning they share an endpoint in G . This lets us trace a closed walk in G that uses every edge exactly once. We must also check that every vertex of G is visited: since every edge appears and every vertex (with nonzero degree) is an endpoint of some edge, all vertices are covered. Hence G is Eulerian. $\square$

- (a) Prove that every graph G with n vertices and with all vertices having degree at least $\frac{n}{2}$ has a Hamiltonian cycle.
- (b) Is it sufficient to assume degrees at least $\lfloor n/2 \rfloor$ in part (a)?

 Remarks

Strategy for (a). Consider a *longest path* $P = v_1 v_2 \cdots v_t$ in G .

- First show that $t = n$ (the path visits every vertex).
- Then show that v_1 and v_t have a common “rotation” index: if v_1 is adjacent to v_k , then check whether v_t is adjacent to v_{k-1} . Use a **pigeonhole** / counting argument on the neighbors of v_1 and v_t among $\{v_1, \dots, v_t\}$.

For (b): Try small cases. What happens when n is odd?

Answer: Part (a): Proof of Dirac's Theorem.

Let G have n vertices and minimum degree $\delta(G) \geq n/2$.

Step 1: G is connected. If u and v were in different components, the component of u has at most $n - 1$ vertices, so $\deg(u) \leq n - 2$. But the component of u has $\geq \deg(u) + 1 \geq n/2 + 1$ vertices, and similarly for v , giving $\geq n + 2$ vertices total — contradiction. (More simply: any two vertices share a common neighbor by pigeonhole, since $|N(u)| + |N(v)| \geq n$ but there are only $n - 2$ other vertices.)

Step 2: A longest path spans all of G . Let $P = v_1 v_2 \cdots v_t$ be a longest path. All neighbors of v_1 lie on P (otherwise we could extend P). Similarly for v_t . So $\deg(v_1) \leq t - 1$ and $\deg(v_t) \leq t - 1$. But $\deg(v_1) \geq n/2$, so $t \geq n/2 + 1$.

Claim: $t = n$. Suppose $t < n$. Then there exists a vertex $w \notin V(P)$. Since G is connected, w is adjacent to some v_i on P . But then $w, v_i, v_{i-1}, \dots, v_1$ or $w, v_i, v_{i+1}, \dots, v_t$ extends to a longer path only if v_1 or v_t has a neighbor off the path. We refine: first close P into a cycle (Step 3), then use w to find a longer path, contradiction.

Step 3: Close the path into a Hamiltonian cycle. Define two sets of indices in $\{2, 3, \dots, t\}$:

$$A = \{k : v_1 \sim v_k\},$$

$$B = \{k : v_t \sim v_{k-1}\}.$$

Then $|A| \geq n/2$ and $|B| \geq n/2$, both subsets of $\{2, \dots, t\}$ which has $t - 1$ elements. If $t < n$, then $|A| + |B| \geq n > t - 1$ (since $t \leq n - 1$), so $A \cap B \neq \emptyset$ by pigeonhole. Pick $k \in A \cap B$: then $v_1 \sim v_k$ and $v_t \sim v_{k-1}$. The cycle

$$v_1, v_2, \dots, v_{k-1}, v_t, v_{t-1}, \dots, v_k, v_1$$

has length t . Since there is a vertex $w \notin V(P)$ adjacent to some vertex on this cycle (by connectedness), we can open the cycle at that vertex and extend to get a path of length $> t$, contradicting maximality.

If $t = n$, we repeat the pigeonhole argument: $|A| + |B| \geq n$ and both lie in $\{2, \dots, n\}$ of size $n - 1$, so $A \cap B \neq \emptyset$. Pick $k \in A \cap B$: the cycle $v_1, v_2, \dots, v_{k-1}, v_t, v_{t-1}, \dots, v_k, v_1$ visits all n vertices — a Hamiltonian cycle. $\square$

Part (b): No, $\lfloor n/2 \rfloor$ does not suffice.

Counterexample for odd n . Take $n = 2m + 1$ (odd) and form the graph $K_{m+1} \sqcup K_{m+1}$ sharing a single vertex v (i.e. two copies of K_{m+1} glued at one vertex). This graph has $2m + 1 = n$ vertices. Every vertex except v has degree $m = \lfloor n/2 \rfloor$, and v has degree $2m$. But the graph is not 2-connected (v is a cut vertex), so it cannot have a Hamiltonian cycle.

Why Dirac's bound is tight: For even n , $\lfloor n/2 \rfloor = n/2$, so the theorem applies. For odd n , the threshold $n/2$ is not an integer, so Dirac requires $\deg \geq \lceil n/2 \rceil$, which is strictly more than $\lfloor n/2 \rfloor$. The counterexample shows this gap matters. $\square$